Week 6: Causality

Empirical Research Methods

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Learning Objectives

- Understanding the difference between correlation and causation
- Learn to identify different terms of causal issues
- Visualize the causal network around your research problem
- Identify causal links that could harm ability to identify and isolate relationships of interest
- Connect causal theory to practical applications

Correlation vs Causality

Correlation

More generally known as **association**, it is the relationship between two variables, such that a change in one corresponds to a consistent change in the other.

Causation

The *confirmed* knowledge of a direct relationship, such that purposefully changing one will result in a corresponding and consistent change in the other.

The difference? Causation is the understanding that a mechanistic relationship exists, whereas correlation/association is the knowledge that two variables tend to move together in a certain manner.

Corr. v Cause: Examples

Do the examples below share a causal relationship, an associative relationship, or are they unrelated?

1. Atmospheric greenhouse gas and global average temperature
2. A stock price and the revenue of a company
3. Kerosene used in Cuba and Google searches for 'CIA hotline'

Corr. v Cause: Examples

Answers from the previous slide:

1. **Causal relationship**, see any IPCC report
2. **Usually correlated**, though not always
3. **Correlated**

Causal Effects and Counterfactuals

Counterfactual: The outcome that would have occurred had the treatment been different from what was actually received.

Causal Effect: The effect caused by making a specific change in the explanatory variable. This is the gap between the value of the metric of interest between the observed reality and the counterfactual scenario, which we can never see!

How can we then try to estimate the causal effect?

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How can we then try to estimate the causal effect?

Answer

By having similar observational units experience different treatments.

But how do we define "similar?"

Identifying Causality: Possible Issues

Confounder

A variable with jointly determines both the target variable and the explanatory variable of interest.

Reverse Causality

Swapped understanding of which variable is the explanatory variable and which is the dependent variable in a causal relationship.

Collider

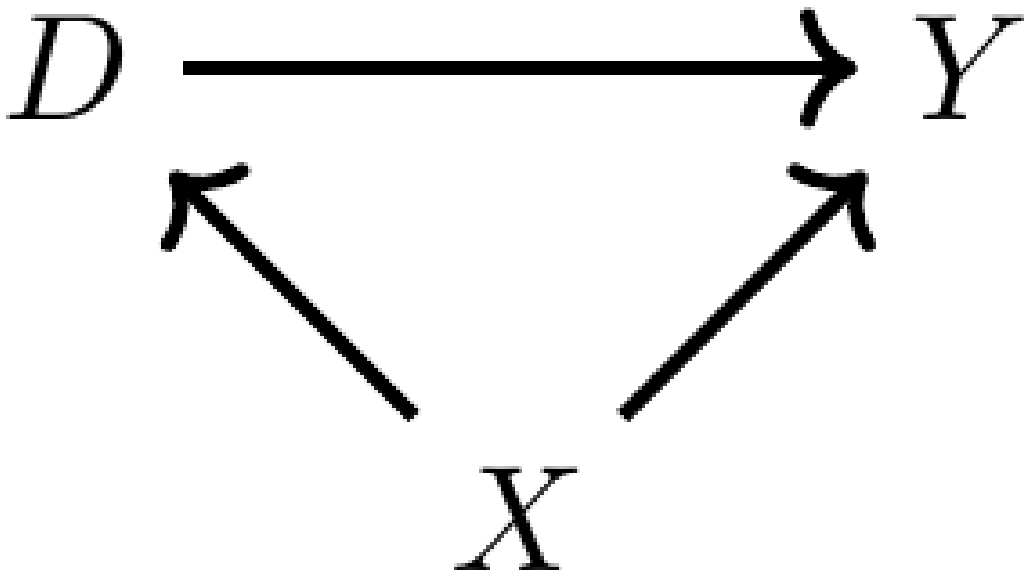
A variable is the causal product of two other variables and we control for i, inducing spurious correlation from the other!

How can we identify these difficult patterns when thinking about such a hectic space with all of our different variables?

Directed Acyclic Graphs

DAG: Graphical visualization of the variables involved in the experiment

- **Nodes**: The variables themselves
- **Arrows**: The causal effect and its direction
- **Paths**: A connection of arrows that can describe the relationship between two variables



DAG Terminology

For point i, point j has the following relation to it under the conditions below.

- **Ancestor**: There exists a directed path back from j to i
- **Parent**: There is a single directed link from j to i
- **Child**: There is a single directed link from i to j
- **Descendant**: There exists a directed path back from i to j
- **Mediator**: j exists on a directed path between i and another variable

Heinze-Deml 2021a

Understanding Backdoor Pathways

Backdoor pathway: A route causally connecting the explanatory and dependent variable that does not originate with explanatory variable Peters, Janzing, and Schölkopf 2017.

Warning!

Backdoor pathways induce additional causal effects that you don't want to estimate as a part of your study. Therefore, you must block all the backdoor pathways.

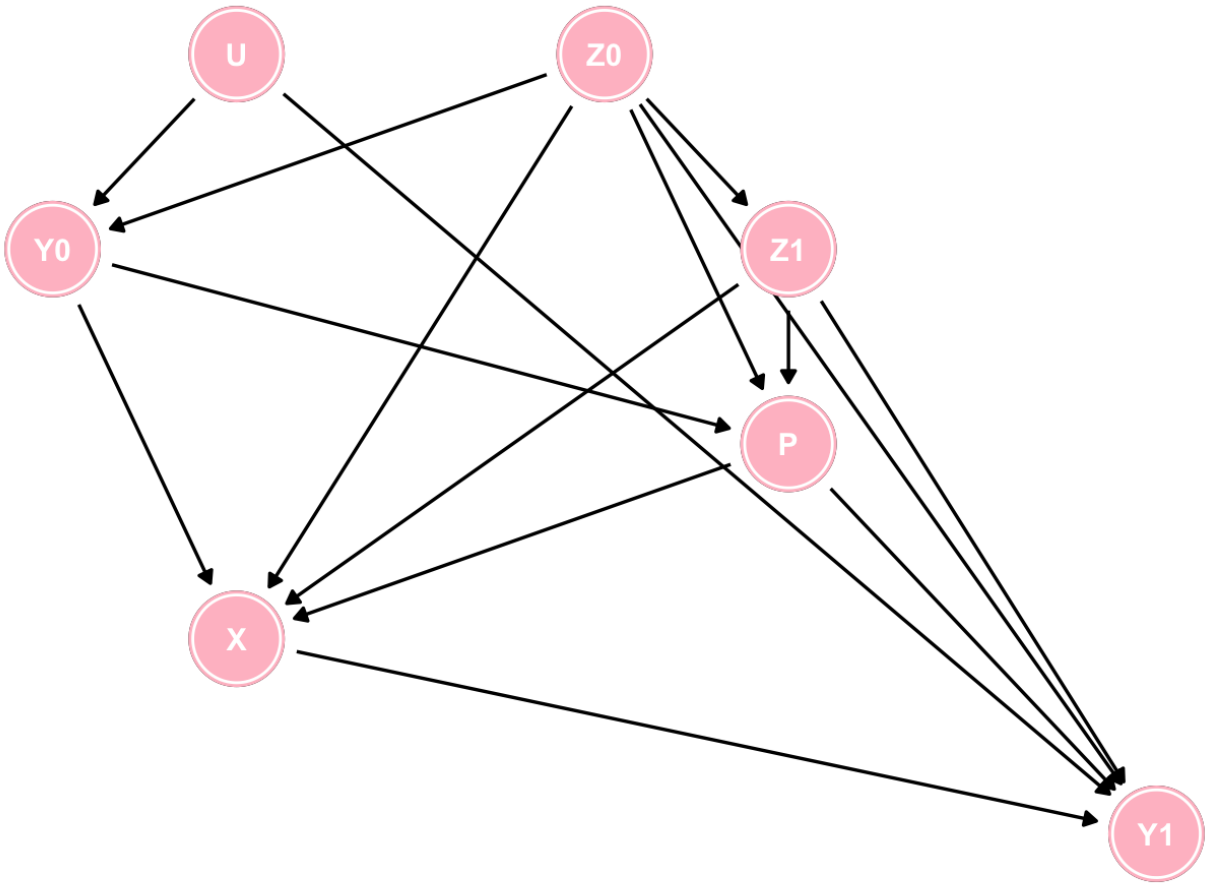
Blocking a pathway is the process by which we control the values of the variables which also have an impact on the explanatory variable (and also the treatment variable.

Controlling backdoor pathways

Identifying which pathways need to be blocked can be done by

1. Draw your DAG with all variables impacting one another, not just those targeted by the study.
2. Remove the direct causal path between the explanatory variable and the dependent variable
3. Find all paths where the aforementioned variables are linked by a common parent and include these in your set of variables to control. *Be careful not to induce correlation via making a collider after controlling!*

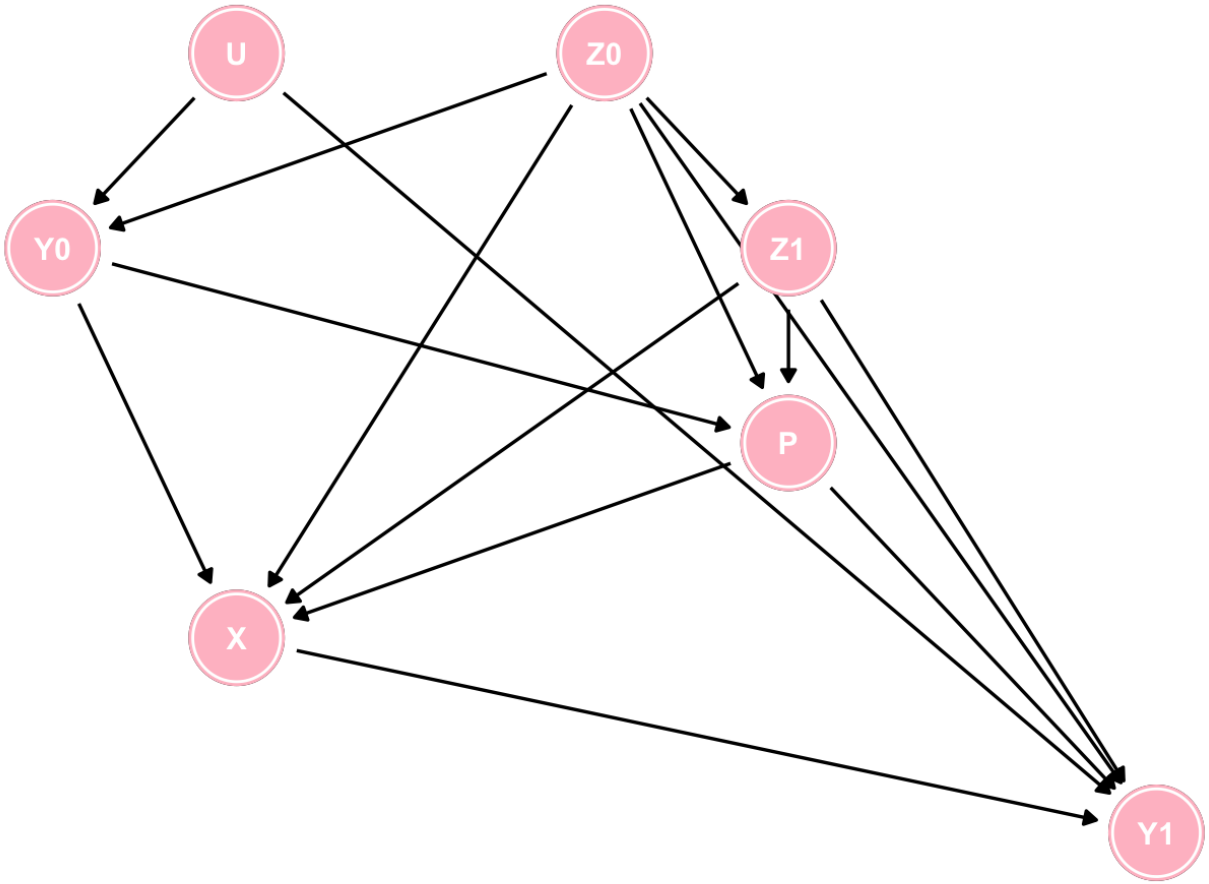
DAGs: An Example



Q1: What are the ancestors of X?

Q2: Is the set $\{P, U, Z_0, Z_1\}$ enough to block all backdoor paths between X and Y_1?

DAG: Example Answers



A1: Y_0, U, Z_0, Z_1, P

A2: Yes, as well as P, Y_0, Z_0, Z_1. Why is this one also okay?

Backdoor Pathways: Controller vs Collider

	Confounder	Collider
Main attribute	common cause	common effect
Association	contributes to the association between its effects	does not contribute to the association between its causes
Type of path	open path	blocked path
Effect of conditioning	blocked path	open path
Bias before conditioning?	Yes, confounding bias	No
Bias after conditioning?	No	Yes, colliding bias

Other Biases Impacting Causality

Selection Bias

Bias which results from experimental units not following the treatment they were assigned to.

- Compliers: Follow assigned treatments
- Defiers: Always do the opposite of the assignment
- Always treated: Always take the treatment
- Never treated: Never take a treatment

You must consider whether or not there is a causal explanation for these behaviours. Return to your DAG!

Using Causal Principles to Design Experiments

What does it mean to control a variable in practice?

And how do we do this in practice?

Using Causal Principles to Design Experiments

What does it mean to control a variable in practice?

Answer

By determining the value of the variable to be controlled.

And how do we do this in practice?

Answer

- In an RCT, by randomizing the treatment variable across different values of the variables to be controlled.
- Matching observational units across the other variables once they are already found to be in the treatment in control groups found outside a laboratory setting.
- Instrumental variables: using a DAG, identify a variable that isolates all the effects of a desired treatment

Recommended Resources

The entirety of ETH's 2021 Causality Lecture from COVID is online. This includes videos, slides, and exercises. Here is the link.

For a social science-oriented resource, try the book Applied Causal Inference with R

Citations and Resources I

- Bauer, Paul C. (May 2020). *12.2 Compliers and non-compliers | Applied Causal Analysis (with R)*. URL: https://bookdown.org/paul/applied-causal-analysis/compliers.html (visited on 04/01/2026).
- Brophy, James (2021). *Chapter 7 Causal inference & directed acyclic diagrams (DAGs) | (Mostly Clinical) Epidemiology with R*. URL: https://bookdown.org/jbrophy115/bookdown-clinepi/causal.html (visited on 03/31/2026).
- Cunningham, Scott (Jan. 2021). *Causal Inference The Mixtape*. ISBN: 978-0-300-25168-5. URL: https://mixtape.scunning.com/ (visited on 03/23/2026).
- Heinze-Deml, Christina (Mar. 2021a). *02_DAGs_0303.pdf*. en. URL: https://polybox.ethz.ch/index.php/s/wvMg0qFkoLnF5RP (visited on 03/31/2026).

Citations and Resources II

- Heinze-Deml, Christina (Mar. 2021b). *03_CausalDAGs_1003.pdf*. en. URL: https://polybox.ethz.ch/index.php/s/r92EmTHhEkccT07 (visited on 03/31/2026).
- Peters, Jonas, Dominik Janzing, and Bernhard Schölkopf (Nov. 2017). *Elements of Causal Inference: Foundations and Learning Algorithms*. en. Ed. by Francis Bach. Adaptive Computation and Machine Learning series. Cambridge, MA, USA: MIT Press. ISBN: 978-0-262-03731-0. URL: https://mitpress.mit.edu/9780262037310/elements-of-causal-inference/ (visited on 03/31/2026).

Topics of interest

We will have more bonus lectures this semester following the cancellation of planned external speakers. Do you have any topics or case studies you want to go over?